# Age- and Arm-Stratified Co-expression Network Rewiring in the Murine Kidney During and After Spaceflight

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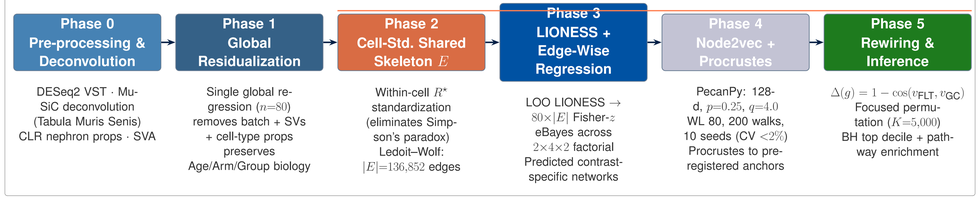
## Analytical Pipeline — node2vec-Driven Transcriptomic Network Rewiring Framework

**Skeleton $E$ fixed across Phases 2–5: same nodes, same edges, model-based weights only**




## Abstract

Astronauts exhibit elevated nephrolithiasis risk post-flight, with prior work linking microgravity to distal nephron remodeling and renal transporter dysregulation. Whether spaceflight alters *regulatory coordination* among genes—shifting co-expression architecture without necessarily changing mean expression—remains uncharacterised.

We analysed OSD-771 (RRRM-2), bulk kidney RNA-seq from **80 female C57BL/6NTac mice** under a 2×4×2 factorial design crossing **Age** (Young 16 wks; Old 34 wks), **Arm** (ISS-T; LAR), and **Environment Group** (FLT, GC, VIV, BSL). MuSiC deconvolution removed compositional confounding. Sample-specific networks via LIONESS on a shared sparse skeleton, followed by node2vec embeddings aligned by Procrustes, yielded per-gene *rewiring scores* across four primary contrasts.

Focused permutation testing identified **11 significantly rewired genes in Young ISS-T** (FDR $< 0.05$), including *Slc6a20a*, *Cry2*, and *Gc*, with two additional hits in Old ISS-T (*Prlr*, *Nr4a1*). Pathway enrichment revealed ECM/complement remodeling in Young ISS-T, EMT/steroid biosynthesis in Old ISS-T, and **retinoic acid metabolism** as the most robust cross-arm signal (OR = 19.9, LAR Young). Rewiring was statistically orthogonal to DE ($\rho \approx 0.03$).

## Background & Rationale

- Astronauts experience ∼5× elevated kidney stone risk post-flight; DCT ion-transport dysregulation is implicated.
- Standard DE tests only *mean shifts*; regulatory *rewiring* at stable mean expression is invisible to conventional analyses.
- RRRM-2 (OSD-771): largest factorial kidney RNA-seq spaceflight dataset, $n$=80.
- **Central question:** Does spaceflight restructure the kidney co-expression network in an age- and arm-dependent manner?

**Dataset — NASA OSD-771 (RRRM-2)**

| Factor | Levels |
|---|---|
| Age | Young (16 wks), Old (34 wks) |
| Arm | ISS-T (terminal), LAR (live return) |
| Environ. Group | FLT, GC, VIV, BSL |
| Per cell | 5 mice / age |

## Results: Rewiring Landscape

Mean rewiring ($\Delta$) was similar across all four contrasts (0.269–0.284), indicating a **genome-wide perturbation**. Maximum single-gene rewiring reached 0.547—≈ $twicethegenome - wideaverage^-identifyingindividualhubgenesasextremeou$

**Key result:** Spearman $\rho = 0.02$–$0.05$ between $\Delta$ and $|\log_2 \mathrm{FC}|$ ($R^2 < 0.3\%$) across all contrasts. **Rewiring is statistically orthogonal to differential expression.**

*[Figure: 4-panel scatter — $\Delta$ vs. $|\log_2 FC|$]*
*One panel per contrast. Spearman $\rho$ annotated. Focused-permutation hits (FDR < 0.05) coloured.*

## Results: Focused Permutation Hits (FDR $< 0.05$)

| Contrast | Gene | $q$ (BH) | Function / Biological relevance |
|---|---|---|---|
| ISS-T Young | *Slc6a20a* | 0.025 | Renal proline/glycine transporter (proximal tubule) |
| ISS-T Young | *Trf* | 0.025 | Transferrin; iron and protein trafficking hub |
| ISS-T Young | *Fabp7* | 0.033 | Fatty acid-binding protein; lipid metabolism reprogramming |
| ISS-T Young | *Azgp1* | 0.037 | Zinc-alpha-2-glycoprotein; lipid mobilization and metabolic adaptation |
| ISS-T Young | *Foxq1* | 0.043 | Forkhead TF; epithelial differentiation and polarity |
| ISS-T Young | *Slc22a28* | 0.043 | Organic anion transporter (proximal tubule) |
| ISS-T Young | *Adgrg4* | 0.044 | Adhesion GPCR; urogenital epithelium |
| ISS-T Young | *Gc* | 0.044 | Vitamin D-binding protein; mineral homeostasis |
| ISS-T Young | *Pfkp* | 0.050 | Platelet-type PFK; rate-limiting glycolytic enzyme |
| ISS-T Young | *Cry2* | 0.050 | Circadian repressor; 16 sunrises per day on ISS |
| ISS-T Young | ENSMUSG00000043293 | 0.043 | Unmapped locus |
| ISS-T Old | *Prlr* | 0.025 | Prolactin receptor; renal osmoregulation |
| ISS-T Old | *Nr4a1* | 0.025 | Nur77; apoptosis and renal injury response |
| LAR Young | *Eps8l1* | 0.050 | Actin dynamics regulator; tubular morphology |

*[Figure: Rank plots for all 4 contrasts]*
*Genes ranked by $\Delta$; FDR < 0.05 hits labelled. ISS-T Young shows the highest density of significant hits.*

## Results: Pathway Enrichment (FDR $< 0.05$)

| Contrast | Pathway (Database) | OR | $q$-value | Hits |
|---|---|---|---|---|
| ISS-T Young | ECM Organization (Reactome) | 3.9 | 0.048 | 13 |
| ISS-T Young | Collagen Chain Trimerisation (Reactome) | 13.5 | 0.048 | 5 |
| ISS-T Young | Complement and Coagulation (KEGG) | 9.5 | 0.007 | 8 |
| ISS-T Young | Protein Digestion and Absorption (KEGG) | 10.2 | 0.007 | 8 |
| ISS-T Old | Epithelial–Mesenchymal Transition (Hallmark) | 4.6 | 0.039 | 12 |
| ISS-T Old | Steroid Hormone Biosynthesis (KEGG) | 13.1 | 0.039 | 6 |
| LAR Young | **Retinol Metabolism (KEGG)** | **19.9** | $1.6 \times 10^{-5}$ | **9** |
| LAR Young | Xenobiotics/CYP450 (KEGG) | 12.4 | 0.001 | 8 |
| LAR Young | Complement and Coagulation (KEGG) | 8.1 | 0.026 | 7 |
| LAR Old | NAFLD / Complex I subunits (KEGG) | 6.2 | 0.001 | 13 |
| LAR Old | ECM Organization (Reactome) | 4.9 | 0.001 | 16 |

*[Figure: Pathway bubble dot plot]*
*Axes: pathway × contrast. Bubble size = hit count. Colour = $-\log_{10}(q)$. Retinol metabolism prominent.*

## Results: ISS-T vs. LAR and Convergent Retinoic Acid Signal

ISS-T shows **greater and more significant rewiring** than LAR across all contrasts (11 hits vs. 1 in Young; 2 vs. 0 in Old), consistent with *partial network recovery* following return to terrestrial gravity.

- ∼**15%** top-decile genes are **ISS-T-specific** (ephemeral; resolved on return): *Cry2*, *Foxq1*, renal transporters.
- ∼**22%** are **persistent** (rewired in both arms): *Cfh*, *Vwf*, *Col1a1*, *Fn1*.

**Convergent signal: Retinoic acid metabolism** is the single most reproducible finding. OR = 19.9 in LAR Young ($q$ =